

1 Rescorla Wagner (wikipedia is a good source)

$$\Delta V_X^{n+1} = \alpha_X \beta (\lambda - V_{\text{tot}})$$

and

$$V_X^{n+1} = V_X^n + \Delta V_X^{n+1}$$

where

- ΔV_X is the change in the strength, on a single trial, of the association between the CS labelled "X" and the US
- α is the salience of X (bounded by 0 and 1)
- β is the rate parameter for the US (bounded by 0 and 1), sometimes called its association value
- λ is the maximum conditioning possible for the US
- V_X is the current associative strength of X
- V_{tot} is the total associative strength of all stimuli present, that is, X plus any others